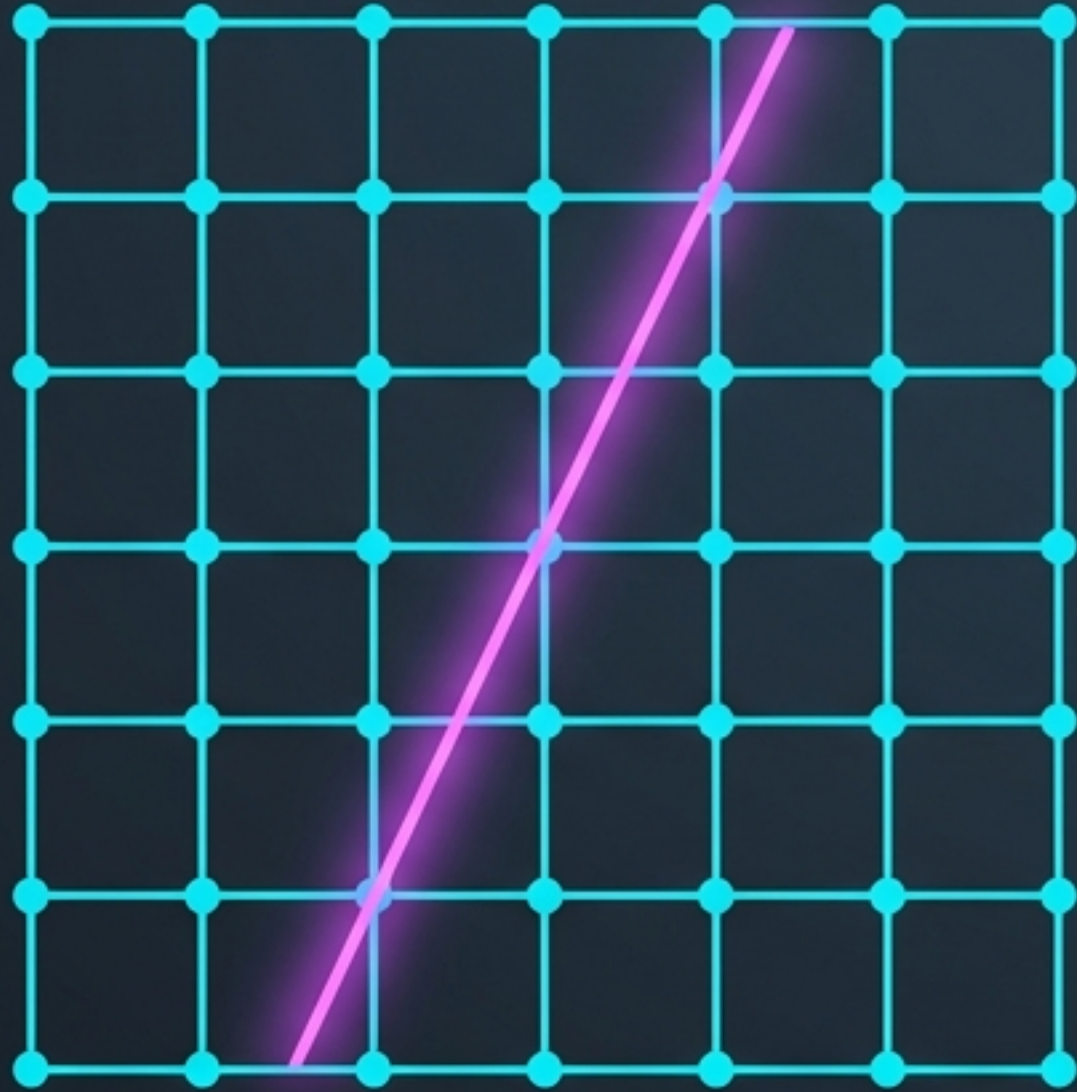


$$\left(\frac{3x^3 - 7x^2 + 2x}{x - 2} \right) = \begin{matrix} & 3x^2 & - & x \\ & \swarrow & & \searrow \\ & \dots, 0, 2, 10, \dots \end{matrix}$$

The End of a 126-Year Mystery

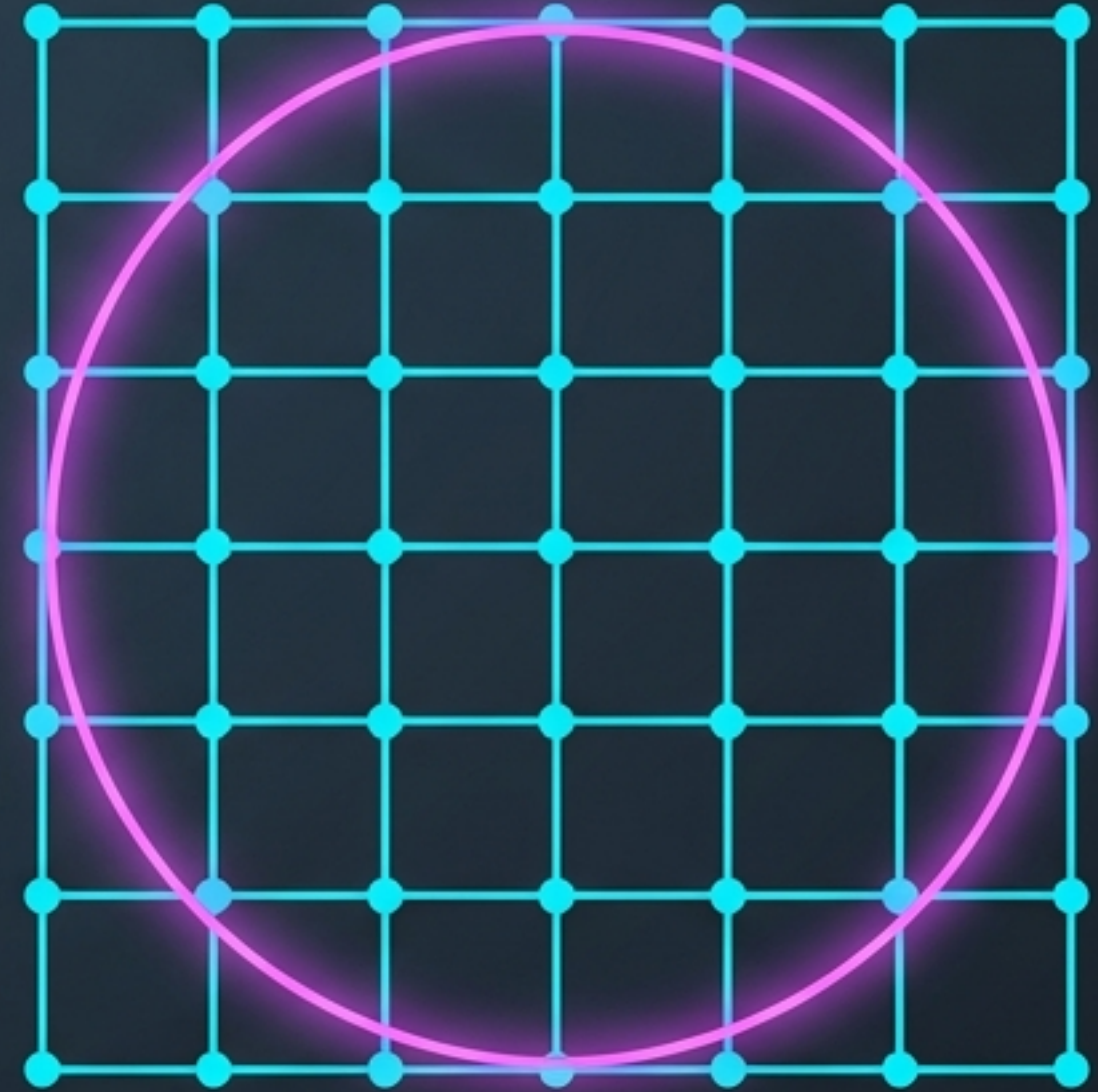
How Artificial Intelligence Settled Hilbert's Tenth Problem for the Rationals

$$2x = 1$$



Has a rational solution ($x = 1/2$)
but NO integer solution.

$$x^2 + y^2 = 3$$



Has NO rational or integer solutions.

A Diophantine equation is simply a polynomial where we are only allowed to ask one question: Are there solutions inside a specific mathematical universe?

The Flashlight Problem

Searching confirms presence, but never absence. To truly solve Hilbert's problem, an algorithm must be able to confidently stop and say NO. In logic, an inability to guarantee a stop is known as the Halting Problem.

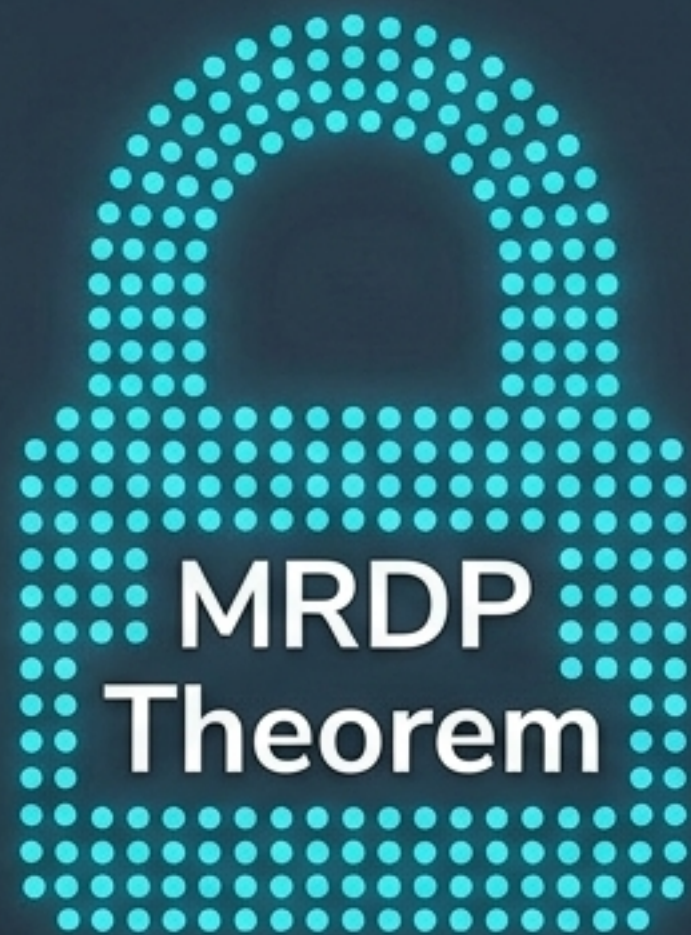


If a solution exists, searching
1, 2, 3... will eventually find it.



If a solution does
NOT exist, you
search to infinity.





1900: David Hilbert
poses his 10th Problem.



1961: Davis, Putnam,
and Robinson (DPR)
reduce the problem.

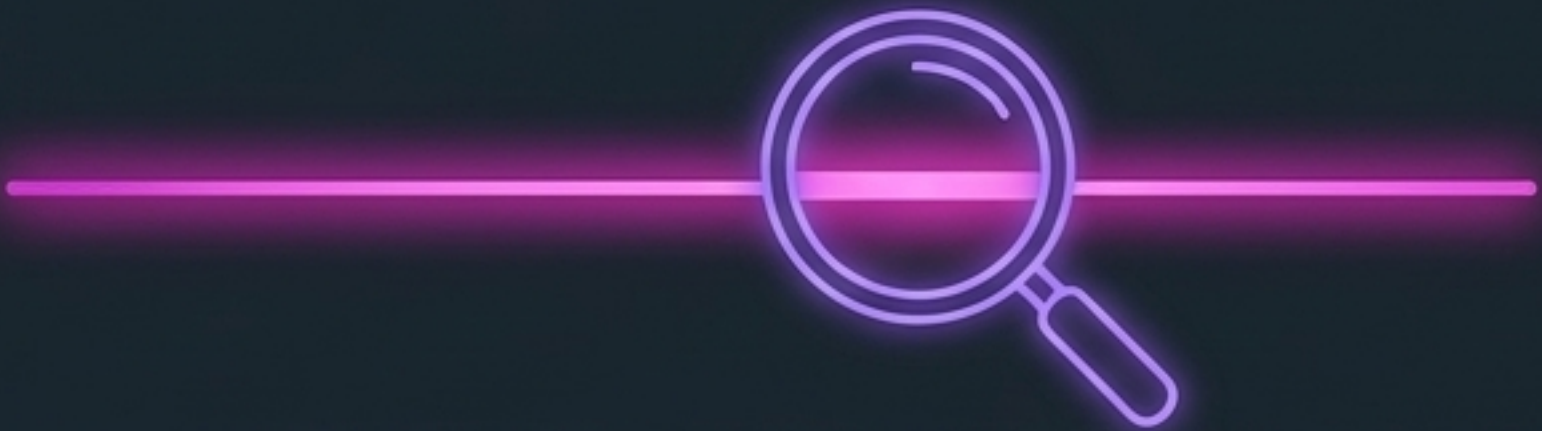


1970: Yuri Matiyasevich
delivers the final blow.



The Integers ($\mathbb{Z}$) are fundamentally undecidable. No algorithm can ever be built that always tells you if an integer solution exists. The first half of Hilbert's dream is dead.

The $\mathbb{Z}$ vs $\mathbb{Q}$ Matrix

Integers $\mathbb{Z}$	Geometry	Rationals $\mathbb{Q}$
		
The Logic Trap		
Diophantine equations only allow “There Exists” statements.		Previous attempts to capture integers inside rationals (Robinson 1949, Poonen 2009, Koenigsmann 2016) all required “For All” statements, breaking the rules of the game.

Mazur’s Conjecture suggests that the smooth, dense geometry of rationals makes a simple, rule-abiding definition of integers geometrically impossible.



Input:

Number of variables is part of the input.

Theorem 1.1:
There is NO algorithm
if a polynomial has a
rational zero.

Corollary 7.1b:

Degree Bound
(A maximum polynomial degree of 4 is sufficient to break mathematics).

Corollary 7.1a

Turing Degree 0'
(Identical in unsolvability to the Halting Problem itself).

The Logic Race: Integers vs. Rationals

Program 1: Search for an Integer Solution.



Program 1: Search for an Integer Solution.

IF AND ONLY IF

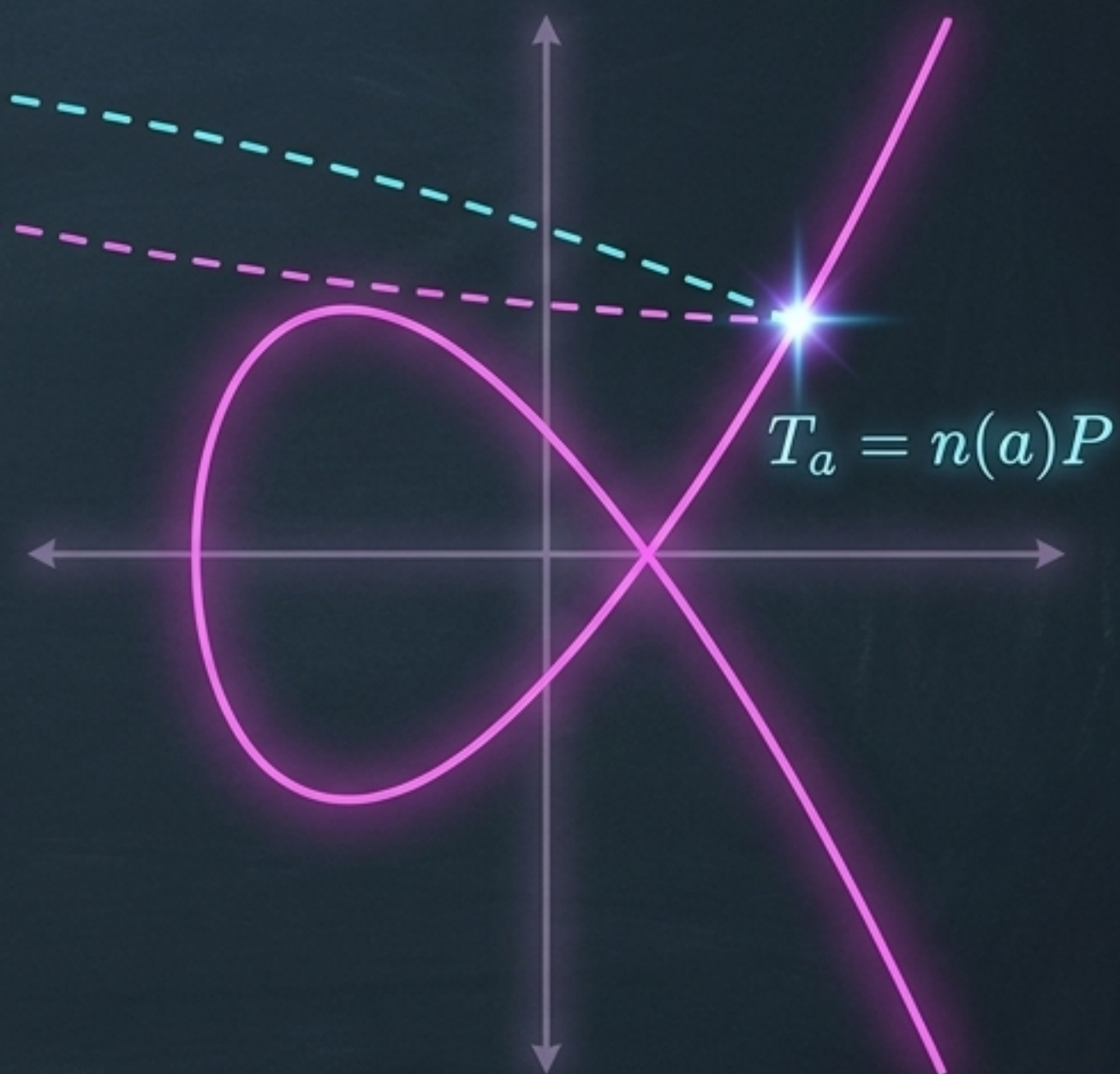
Program 2: Run Finite Rational Tests searching for a Failure.

Program 2: Run Finite Rational Tests searching for a Failure.

The master trap: An integer solution exists if and only if every single rational test succeeds. Therefore, one of these programs MUST eventually stop. If we had an algorithm for Q , we could run this race and solve the integer problem—but the 1970 MRDP theorem proved integers are unsolvable! Contradiction achieved.

$$y^2 = x^3 - d^2x$$

over $\mathbb{Q}(\sqrt{2})$, with $d = 75 - 53\sqrt{2}$



By assuming the impossible (compactness), a fake rational root is trapped in a phantom ring. We use a specific Rank-One Elliptic Curve as a mathematical ruler to measure every element. The mapping preserves addition, but mysteriously fails to preserve multiplication.

The Prime Microscope

x

$$a = n(a) \bmod q^k$$

Original
Elliptic
Curve

Conjugate
Twin

Contact Primes (q)

To fix the multiplication problem, we examine the fractions $\frac{\text{prime}}{\text{prime}}$ by prime.

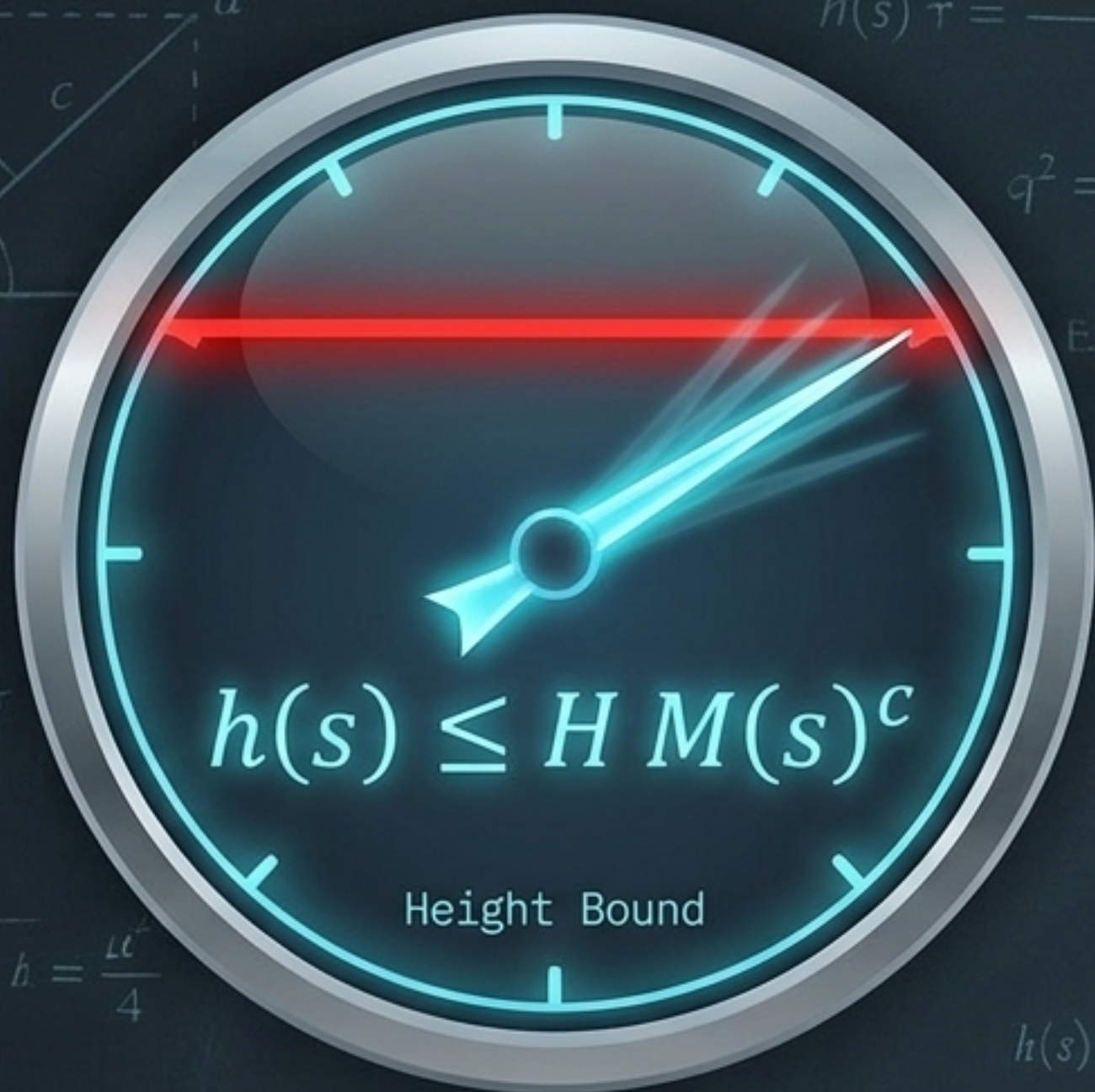
Using a "Truncated Formal Logarithm" on both the curve and its twin, we force the phantom elements to mathematically match up with reality to a massive degree of precision (K).



$$h(s) \tau = \frac{zx^2 - qq^2}{x}$$

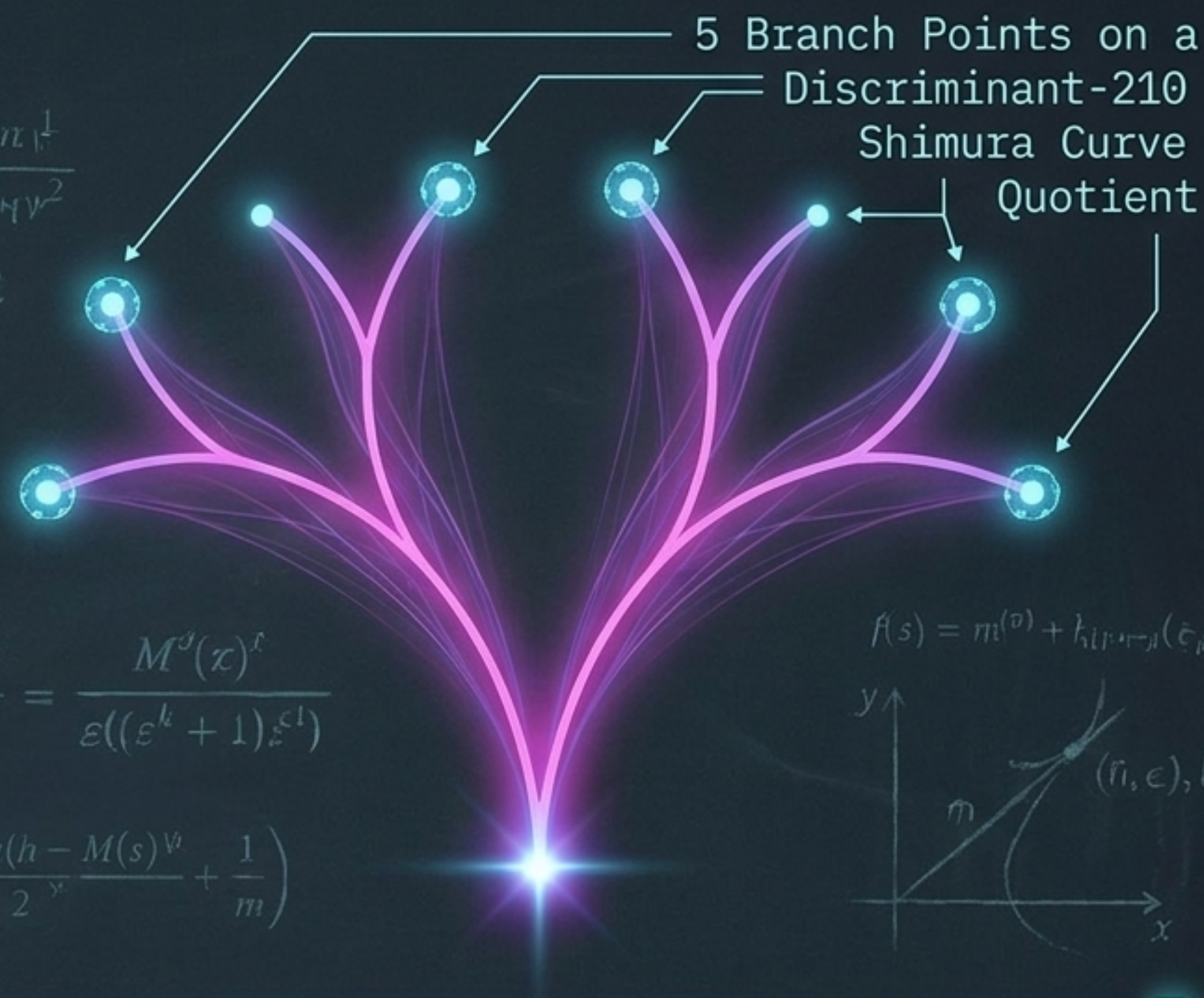
$$q^2 = \frac{n_d}{4n} = \frac{n_d^{\frac{1}{2}}}{4n^{\frac{1}{2}}}$$

$$\text{Exc}, \sigma = 2$$



$$h(s) \leq H M(s)^c$$

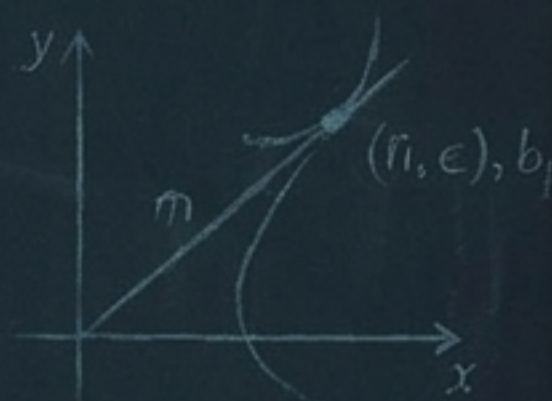
Height Bound



$$\frac{31\epsilon^2}{2} = \frac{M^2(x)^2}{\epsilon((\epsilon^k + 1)\epsilon^1)}$$

$$h(s) = \left(\frac{1}{2h-1} \frac{h(h-M(s)^{1/2})}{s} + \frac{1}{m} \right)$$

$$f(s) = m^{(v)} + h_{1,1,1,1}(\xi_1)$$



We must bound how “complex” or massive these fractions can get (their height). By checking just five fixed points, we can put a hard mathematical ceiling on their size. But there’s a catch: this gauge only works for primes of ODD contact.

The Parity Sieve

Built on a mechanism called the "pointwise 2-converse", the Pole-Parity Formula acts as a strict gatekeeper. It forces the denominators to play by strict parity rules, unlocking the full power of the ODD contact height estimates.

$$P = at^2[1-a, b, c]$$

$$\phi\pi = -\ln^n Z + \left(\frac{r+1}{2}\right)^2$$

$$\bar{V}_i = \left(\frac{1}{z+1}, \dots\right) - \left(\frac{\tau-1}{2}\right)$$

$$= \frac{1}{\sqrt{\pi+3}} \left(\frac{x+b}{2}\right)^2 [-c]$$

$$L_u = w|z|_b + \frac{1}{z^n} \left(\frac{1}{2} \left|1 - 2\ln \frac{1}{(n-b)}\right|^2\right)$$

$$h^2 = h^2 + bh$$

Rational
Possibilities

Height Bounds

C * B

Elliptic Point Growth

$2 * \hat{h}(P) * B^2$

$$\text{For } (x) = e\left(\frac{1}{2\pi}\right) (-b_1 x + 2)$$

$$= 2 \cdot \left(\frac{1}{2\pi}\right) (-2(x(P)^2))$$

$$= +^n \sqrt{\left(1 + \frac{1}{\sqrt{11x^2}}\right)}$$

$$\Gamma_{\text{psul}} = \sqrt{21 - \frac{1}{2\pi}}$$

The Squeeze.

A linear bound cannot contain quadratic growth forever.
The value B is crushed into a tiny, finite box. The
phantom rational root is squeezed until it has no
choice but to be a perfectly whole, discrete Integer.
The trap snaps shut. Contradiction.

$$2 * \hat{h}(P) * B^2$$

The Unopened Doors

No Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ was actually found.

No "Many-One" Reduction.
(The logic map is Turing, not Many-One).

Mazur's topology conjecture remains completely unresolved.

No Variable Bound.
(We don't know the maximum number of variables needed).



The Author

Generated by Artificial Intelligence (OpenAI, Sept 2026).

The Status

Currently a preprint. It awaits rigorous human peer review or **machine formalization** (such as **Lean**) before being permanently etched into mathematical law.

A new era where machines solve the problems that human minds have hunted for a century.

1900: Hilbert's Challenge

1970: The Integer
Defeat (MRDP)

2026: The Parallel Trap
& The Elliptic Squeeze

The Ultimate Takeaway:
The continuous, infinitely divisible
universe of fractions is just as
fundamentally unknowable as the rigid,
discrete universe of integers.

The Undecidable
Continuum (Q)